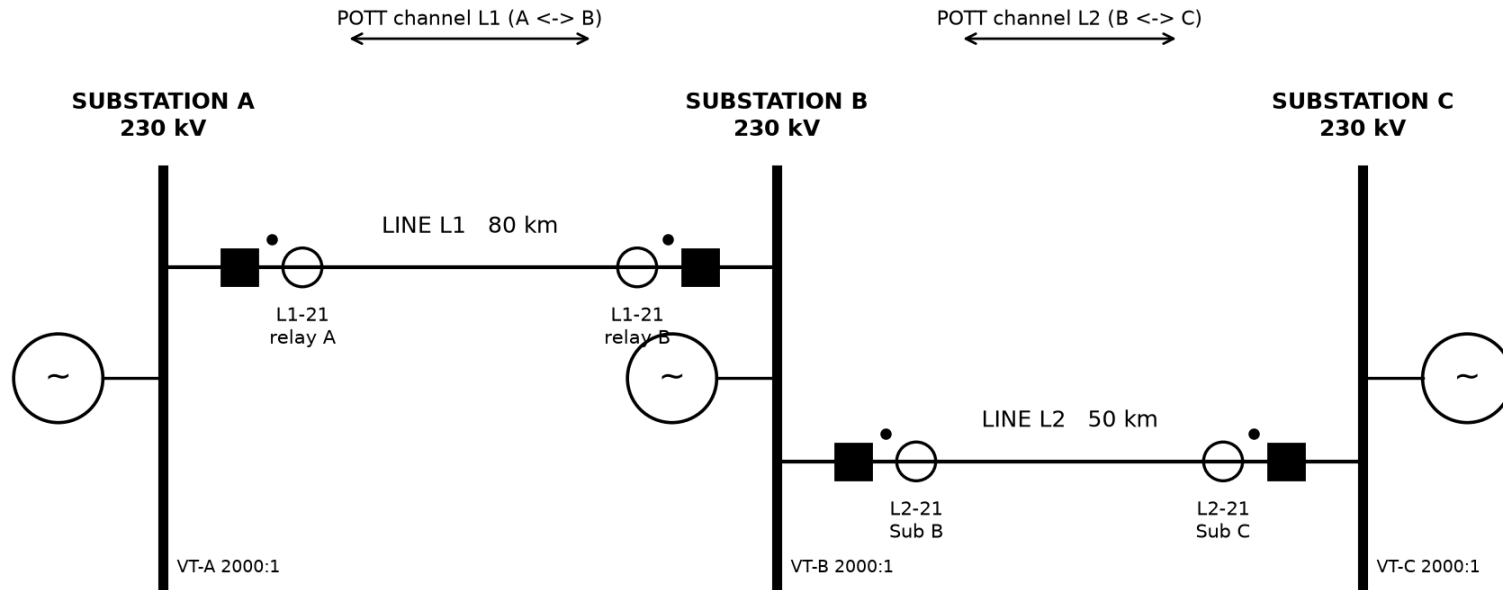


230 kV SYSTEM ONE-LINE (PROTECTION) DWG S-230-001 REV E



Line data (both lines, positive and negative sequence): $z_1 = 0.05 + j0.48$ ohm/km. Line charging negligible. Each line relay takes its voltages from its substation's 230 kV bus VT (VT-A, VT-B, VT-C), so the two relays at Substation B see the same bus voltages. Filled square = closed breaker. CT symbol: circle with polarity dot (P1). Relay current convention: a current entering the CT at P1 is measured as positive, i.e. flowing from the bus into the line.